

CAP. 5

SOME UNSUPERVISED LEARNING METHODS

References:

→ from textbook: cap. 10 - Unsupervised Learning

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Text and examples are taken from:

G. James et al., *An Introduction to Statistical Learning: with Applications in R*,
Springer Texts in Statistics, DOI 10.1007/978-1-4614-7138-7 5,
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Chapter 5 - content

- 5.1. Principal components
- 5.2. *K-means*

Obs.: topics are numbered as they appear in the reference book.

Introduction

- have **only a set of features** $X_1, X_2, \dots, X_p$ measured on n observations
- **not interested in prediction**, because we do not have an associated response variable Y
- the **goal** is to discover interesting things about the measurements on $X_1, X_2, \dots, X_p$
 - Is there an informative way to visualize the data?
 - Can we discover subgroups among the variables or among the observations?
- focus on two types of unsupervised learning:
 - **principal components analysis**: for data visualization or data pre-processing before supervised techniques are applied
 - **Clustering**: a broad class of methods for discovering unknown subgroups in data.
- Often performed as part of an **exploratory data analysis**

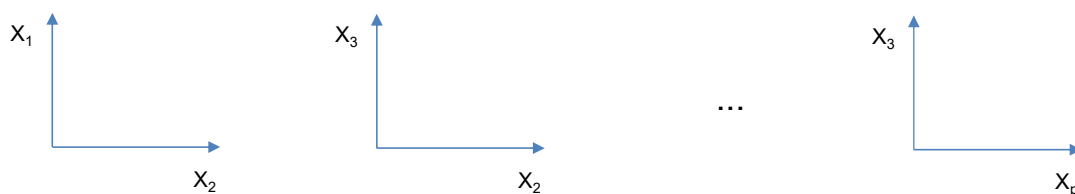
10.2 Principal Components Analysis

- if a large set of correlated variables $\rightarrow$
 - principal components allow us to summarize this set with **a smaller number** of representative variables that collectively **explain most of** the variability in the original set
- To perform principal components regression $\rightarrow$
 - **use principal components as predictors** in a regression model in place of the original larger set of variables.
- **Principal component analysis (PCA)** $\rightarrow$
 - the process by which principal component are computed $\rightarrow$ and the subsequent use of these components in understanding the data
- PCA is an **unsupervised approach**, since it involves only a set of features $X_1, X_2, \dots, X_p$, and no associated response Y

- PCA also serves as a tool for **data visualization** (visualization of the **observations** or visualization of the **variables**).
- focusing on the use of PCA as a tool for unsupervised data exploration

10.2.1 What Are Principal Components?

- Suppose that we wish to visualize n observations with measurements on a set of p features, $X_1, X_2, \dots, X_p$, as part of an exploratory data analysis



- If p is large $\rightarrow$ too many two-dimensional scatterplots;
- probably none of them will be informative $\rightarrow$ just a small fraction of the total information present in the data set.
- we would like **to find a low-dimensional representation** of the data that **captures as much of the information as possible**.

- PCA provides a tool to do just this.
 - It finds a low-dimensional representation of a data set that contains as much as possible of the variation
- PCA seeks a small number of dimensions that are as interesting as possible
 - where the concept of *interesting* is measured by the amount that the observations vary along each dimension.
 - Each of the dimensions found by PCA is a linear combination of the p features.

- Consider a $n \times p$ data set $\mathbf{X}$.
- each of the variables in $\mathbf{X}$ has been **centered** to have mean zero
- the linear combination of the sample feature values of the form

$$z_{i1} = \phi_{11}x_{i1} + \phi_{21}x_{i2} + \dots + \phi_{p1}x_{ip}$$

that has largest sample variance, subject to the constraint

$$\sum_{j=1}^p \phi_{j1}^2 = 1.$$

corresponds to the first principal component Z_1

- We refer to $z_{11}, \dots, z_{n1}$ as the scores of the first principal component.

$\phi_{11}, \dots, \phi_{p1} \longrightarrow$ loadings of the first principal component

- After the **first** principal component Z_1 of the features has been determined, we can find the **second** principal component Z_2 . **Uncorrelated** with Z_1
- The second principal component scores take the form

$$z_{i2} = \phi_{12}x_{i1} + \phi_{22}x_{i2} + \dots + \phi_{p2}x_{ip}$$

$\phi_{12}, \phi_{22}, \dots, \phi_{p2} \longrightarrow$ loadings of the second principal component

Ex.: use of PCA on the USArrests data set

- **50** states in the United States
- number of arrests per 100,000 residents for each of three crimes: **Assault**, **Murder**, and **Rape**.
- Also: **UrbanPop** (the % of the population in each state living in urban areas).
- The PC score vectors have length $n = 50$,
- principal component loading vectors have length $p = 4$.
- PCA was performed after **standardizing** each variable to have mean zero and standard deviation one

	PC1	PC2
Murder	0.5358995	-0.4181809
Assault	0.5831836	-0.1879856
UrbanPop	0.2781909	0.8728062
Rape	0.5434321	0.1673186

TABLE 10.1. The principal component loading vectors ϕ_1 and ϕ_2 for the dataset

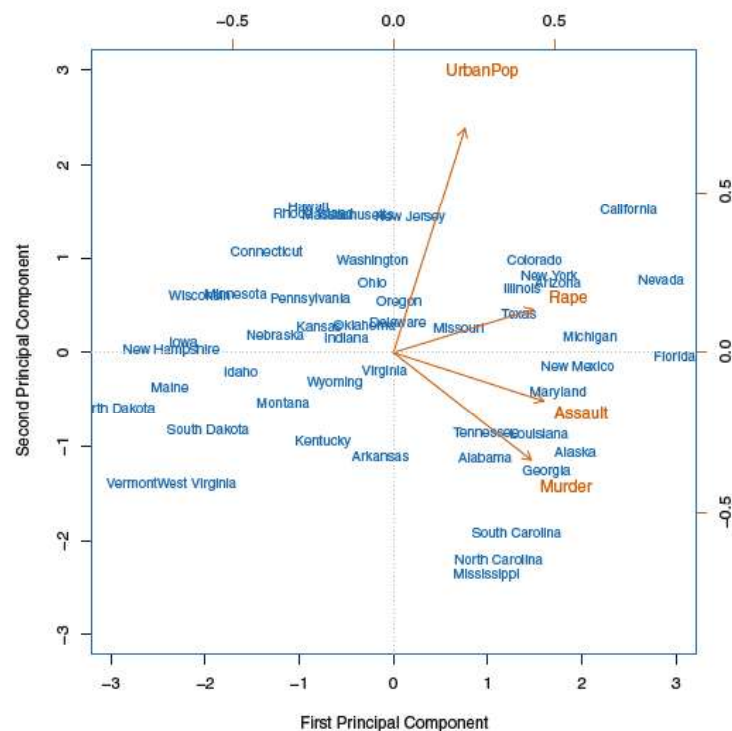


FIGURE 10.1. The first two principal components for the *USArrests* data. The blue state names represent the scores for the first two principal components. The orange arrows indicate the first two principal component loading vectors (with axes on the top and right). For example, the loading for *Rape* on the first component is 0.54, and its loading on the second principal component is 0.17 (the word *Rape* is centered at the point (0.54, 0.17)). This figure is known as a *biplot*, because it displays both the principal component scores and the principal component loadings.

10.2.2 Another Interpretation of Principal Components

- The **first PC loading vector** has a very special property:
 - it is the line in p -dimensional space that is *closest* to the n observations (using average squared Euclidean distance as a measure of closeness). (Figure 6.15)

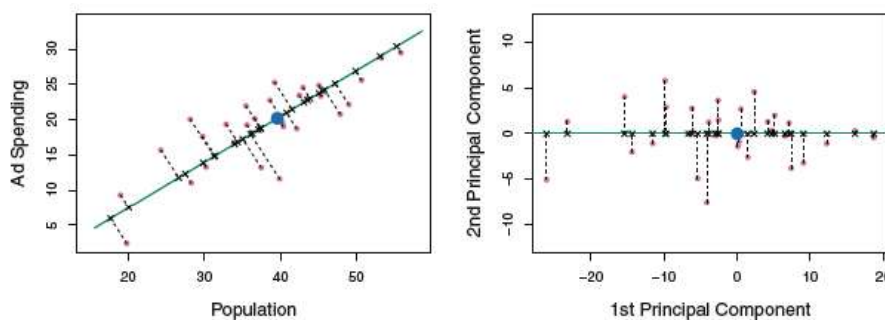


FIGURE 6.15. A subset of the advertising data. The mean *pop* and *ad* budgets are indicated with a blue circle. Left: The first principal component direction is shown in green. It is the dimension along which the data vary the most, and it also defines the line that is closest to all n of the observations. The distances from each observation to the principal component are represented using the black dashed line segments. The blue dot represents $(\overline{\text{pop}}, \overline{\text{ad}})$. Right: The left-hand panel has been rotated so that the first principal component direction coincides with the x -axis.

- the first two PC span the plane that is closest to the n observations (Fig. 10.2)

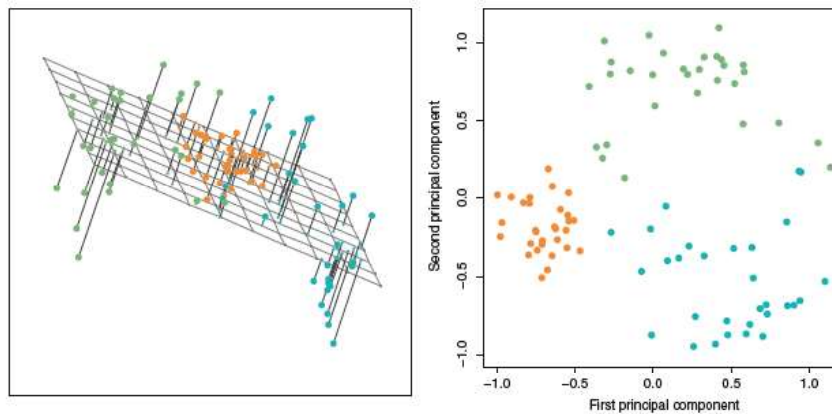


FIGURE 10.2. Ninety observations simulated in three dimensions. Left: the first two principal component directions span the plane that best fits the data. It minimizes the sum of squared distances from each point to the plane. Right: the first two principal component score vectors give the coordinates of the projection of the 90 observations onto the plane. The variance in the plane is maximized.

- The first three PC span the three-dimensional hyperplane that is closest to the n observations, and so forth.

- Using this interpretation, together the first M principal component score vectors and the first M principal component loading vectors provide the best M -dimensional approximation (in terms of Euclidean distance) to the i th observation x_{ij}

$$x_{ij} \approx \sum_{m=1}^M z_{im} \phi_{jm}$$

(assuming the original data matrix $\mathbf{X}$ is column-centered)

- When $M = \min(n - 1, p)$, then the representation is exact:

$$x_{ij} = \sum_{m=1}^M z_{im} \phi_{jm}$$

10.2.3 More on PCA - Scaling the Variables

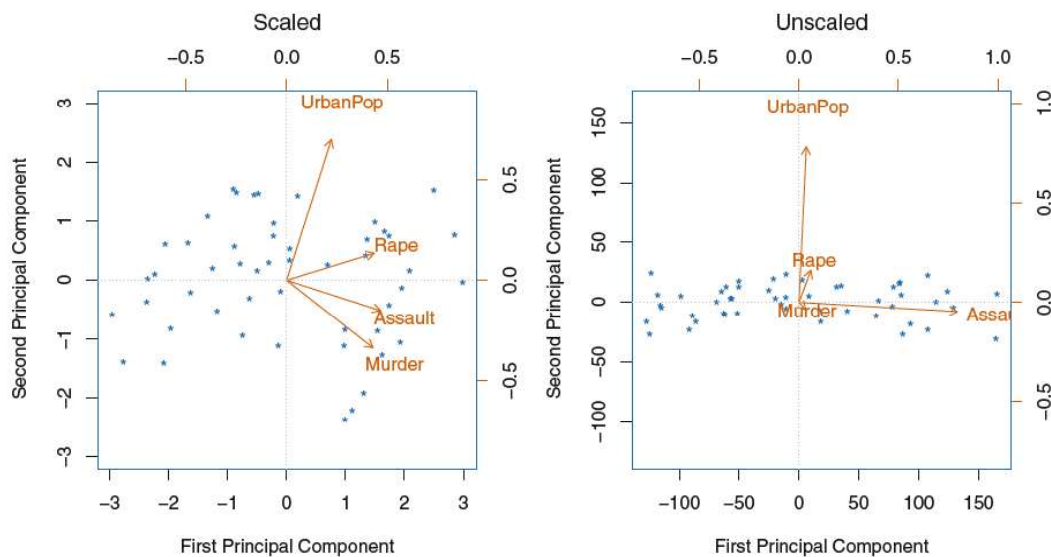


FIGURE 10.3. Two principal component biplots for the **USArrests** data. Left: the same as Figure 10.1, with the variables scaled to have unit standard deviations. Right: principal components using unscaled data. **Assault** has by far the largest loading on the first principal component because it has the highest variance among the four variables. In general, scaling the variables to have standard deviation one is recommended.

10.2.3 More on PCA – The proportion of variance explained – how many PC to use?

- There is no simple rule.

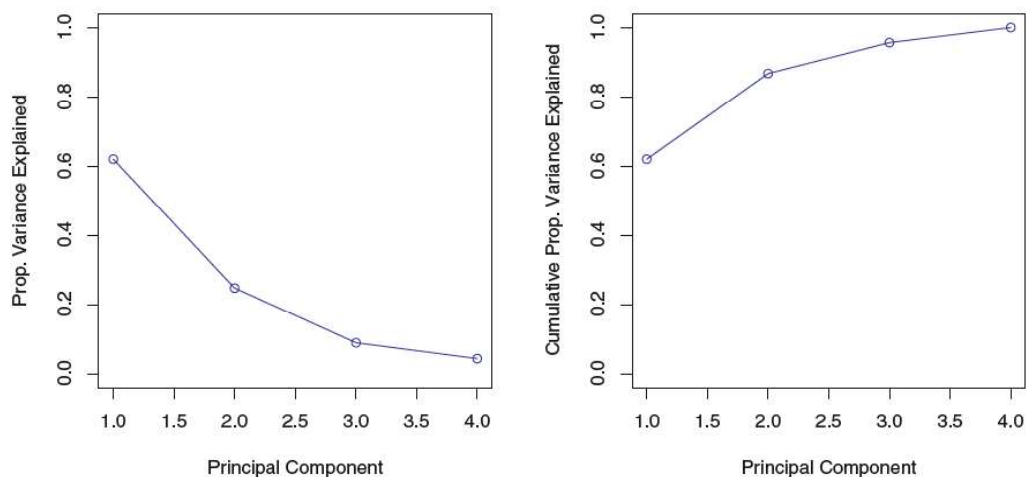


FIGURE 10.4. Left: a scree plot depicting the proportion of variance explained by each of the four principal components in the **USArrests** data. Right: the cumulative proportion of variance explained by the four principal components in the **USArrests** data.

10.3.1 K-Means Clustering

- Must specify K = number of clusters

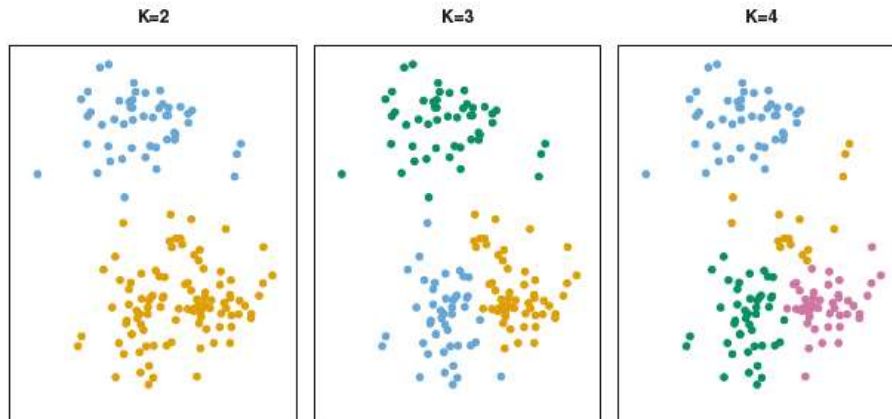


FIGURE 10.5. A simulated data set with 150 observations in two-dimensional space. Panels show the results of applying K-means clustering with different values of K , the number of clusters. The color of each observation indicates the cluster to which it was assigned using the K-means clustering algorithm. Note that there is no ordering of the clusters, so the cluster coloring is arbitrary. These cluster labels were not used in clustering; instead, they are the outputs of the clustering procedure.

- Let $C_1, \dots, C_K$ denote sets containing the indices of the observations in each cluster. These sets satisfy two properties:
 - $C_1 \cup C_2 \cup \dots \cup C_K = \{1, \dots, n\}$. In other words, each observation belongs to at least one of the K clusters.
 - $C_i \cap C_j = \emptyset$ for all $k \neq j$. In other words, the clusters are nonoverlapping: no observation belongs to more than one cluster.

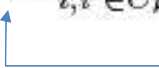
- solve the problem

$$\underset{C_1, \dots, C_K}{\text{minimize}} \left\{ \sum_{k=1}^K W(C_k) \right\}$$

- In words, this formula says that we want to partition the observations into K clusters such that the total within-cluster variation, summed over all K clusters, is as small as possible.

- within-cluster variation
 - most common choice: *squared Euclidean distance*

$$W(C_k) = \frac{1}{|C_k|} \sum_{i, i' \in C_k} \sum_{j=1}^p (x_{ij} - x_{i'j})^2$$


 # of obs in the kth cluster

- In other words, the within-cluster variation for the k th cluster is the sum of all of the pairwise squared Euclidean distances between the observations in the k th cluster, divided by the total number of observations in the k th cluster.

Algorithm 10.1 *K*-Means Clustering

1. Randomly assign a number, from 1 to K , to each of the observations. These serve as initial cluster assignments for the observations.
 2. Iterate until the cluster assignments stop changing:
 - (a) For each of the K clusters, compute the cluster *centroid*. The k th cluster centroid is the vector of the p feature means for the observations in the k th cluster.
 - (b) Assign each observation to the cluster whose centroid is closest (where *closest* is defined using Euclidean distance).
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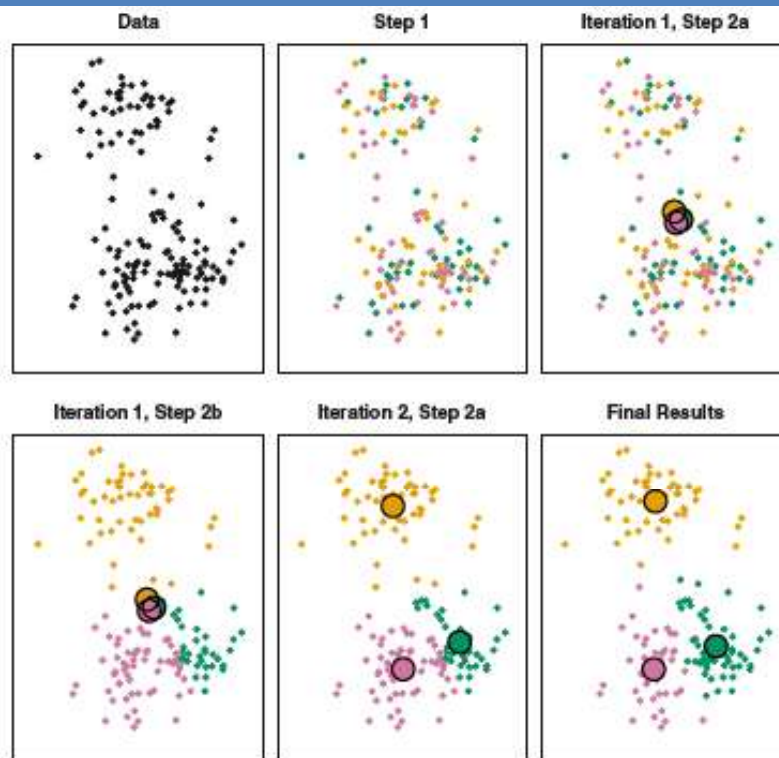


FIGURE 10.6. The progress of the K-means algorithm on the example of Figure 10.5 with $K=3$. Top left: the observations are shown. Top center: in Step 1 of the algorithm, each observation is randomly assigned to a cluster. Top right: in Step 2(a), the cluster centroids are computed. These are shown as large colored disks. Initially the centroids are almost completely overlapping because the initial cluster assignments were chosen at random. Bottom left: in Step 2(b), each observation is assigned to the nearest centroid. Bottom center: Step 2(a) is once again performed, leading to new cluster centroids. Bottom right: the results obtained after ten iterations.

Obs.:

- K-means algorithm finds a local rather than a global optimum → results will depend on the initial (random) cluster assignment of each observation in Step 1 of Algorithm 10.1.
- → run the algorithm multiple times from different random initial configurations.
- Then one selects the best solution, i.e. that for which the objective

$$\underset{C_1, \dots, C_K}{\text{minimize}} \left\{ \sum_{k=1}^K \frac{1}{|C_k|} \sum_{i, i' \in C_k} \sum_{j=1}^P (x_{ij} - x_{i'j})^2 \right\}$$

is smallest

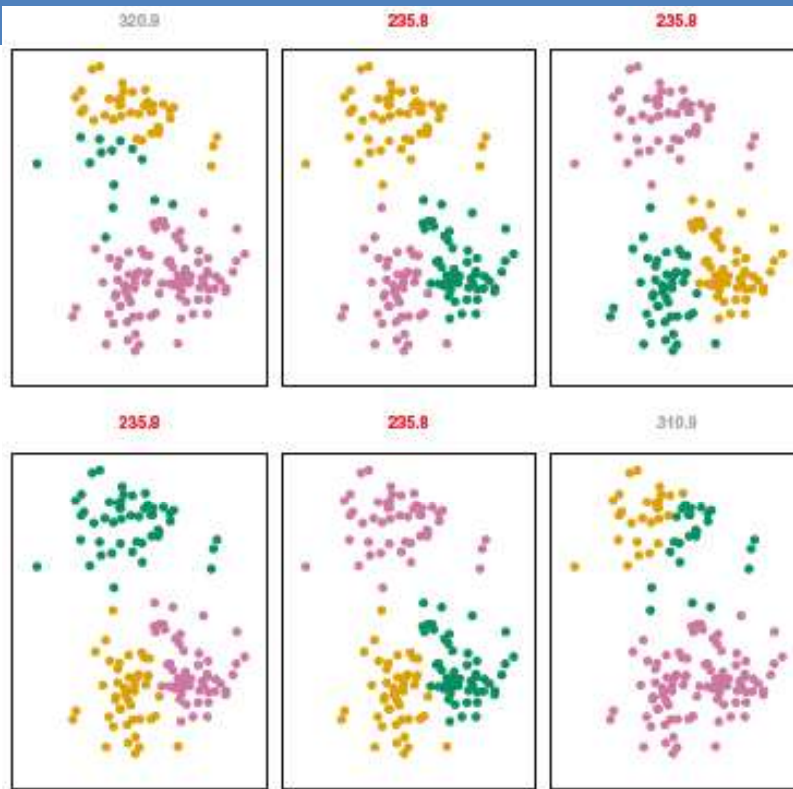


FIGURE 10.7. *K*-means clustering performed six times on the data from Figure 10.5 with $K = 3$, each time with a different random assignment of the observations in Step 1 of the *K*-means algorithm. Above each plot is the value of the objective (10.11). Three different local optima were obtained, one of which resulted in a smaller value of the objective and provides better separation between the clusters. Those labeled in red all achieved the same best solution, with an objective value of 235.8.

10.4 Lab 1: Principal Components Analysis

10.5.1 K-Means Clustering - Lab